



RNN Initialization

UNIVERSIDAD PONTIFICIA COMILLAS
ICAI SCHOOL OF ENGINEERING

Authors	Enrique Campos, Daniela Blanco del Real, Joaquín Mir, Patricia Renart
Study Program	Mathematical Engineering and AI
Submitted on	5th April 2024

1 Project Proposal

1.1 Summary

Recurrent Neural Networks (RNNs) are widely used in various sequential data tasks such as language modelling and sentiment analysis. However, they present significant challenges due to the nature of their architecture. Common problems of RNNs include vanishing gradients, exploding gradients, limited long-term dependency, and, in some cases, the unidirectionality of these models is a challenge. Some, but not all, of these problems could potentially be solved or improved upon with a better initialization.

The state-of-the-art research to solve these training challenges has been in three different direction: algorithmic improvements (like Hessian-Free optimization), neural network architecture (using LSTMs or bidirectional RNNs) and weight initialization

The proposed project will focus on investigating the third point: different types of weight initialization strategies for the RNN and LSTM training and performance.

Through our preliminary research we have identified several initialization techniques that are particularly useful in training RNNs. The initialization of the recurrent weight matrix can be approached in several ways. In their research, Sachin S. Talathi and Aniket Vartak compare IRNN (initialized to Identity matrix), iRNN ($0.01 * \text{Identity matrix}$), nRNN (matrix with all except the highest eigenvalue less than 1), np-RNN (normalized -positive definite matrix with all except the highest eigenvalue less than 1), oRNN (orthogonal random matrix), and gRNN (random Gaussian matrix). This could be a good starting point for our implementation.

We intend to explore other architectures such as Long Short-Term Memory Networks (LSTMs) and initialization methods such as random initialization, Xavier and Kaiming initialization, or orthogonal initialization for their parameters. We aim to test all of these methods and provide meaningful metrics on their effects on performance and training, as well as identifying possible drawbacks and limitations of the different techniques.

1.2 Project plan, models, data ...

This section outlines the planned approach for the project, emphasising the methodologies, models, and tools to be employed.

In order to define our strategy optimally we have chosen to divide the project in three different sections and analyze the necessary tasks to complete each of them.

1.2.1 Initialization strategies on the training and performance of RNN networks

- Theoretical explanation of the different problems and discussion about background research on other initialization papers
- Implementations to elucidate reasons behind the problems
- Feasible solutions

1.2.2 Initialization strategies for LSTM parameters

- Practical implementation with a challenging dataset to explain differences between proposed initializations.
- Graphically illustrate differences in convergence times, computational load, accuracy, etc
- **Data:** Polymorphic music modeling (MIDI files - music21 library)

1.2.3 Most effective initialization strategy for RNNs

- Practical implementation and optimization of various initializations for RNNs
- **Data:** Sentiment Analysis

1.3 Relevant Papers

In this overview, a selection of pertinent papers will be listed. In order to consider the valuable information in each of them, a brief summary of their content will also be provided.

- **Talathi, S. S., and Vartak, A. (2016). Improving Performance of Recurrent Neural Network with ReLU Nonlinearity. Qualcomm Research. San Diego, USA.**

This document tackles the RNN difficulties in training, comparing different types of initialization methods. This paper serves as a starting reference for what our different matrices could look like. For example, it uses IRNN, nRNN or oRNN and compares their performance under the same conditions

[Link to paper](#)

- **Glorot, X., and Bengio, Y. (2010). Understanding the Difficulty of Training Deep Feedforward Neural Networks. DIRO, Université de Montréal, Montréal, Québec, Canada.**

This research highlights issues with random initializations in deep neural network training, such as saturation in top layers with logistic sigmoid activations, slowing learning. Xavier initialization maintains consistent variances across network layers, mitigating vanishing or exploding gradients for improved training dynamics and faster convergence, resulting in superior model performance across tasks and datasets.

[Link to paper](#)

- **Ouannes, P. (2019, March 22). How to initialize deep neural networks? Xavier and Kaiming initialization**

The text discusses Xavier initialization, aiming to maintain activation and gradient variances for stable training. It also introduces Kaiming/He initialization, which considers activation functions unlike Xavier, which approximates the derivative at 0 by 1.

[Link to paper](#)

- **Stephen Merity (June 27, 2016). Explaining and illustrating orthogonal initialization for recurring neural networks**

Orthogonal initialization utilizes matrix eigendecomposition, ensuring stable outcomes through repeated matrix multiplication. By maintaining eigenvalues with absolute values of 1, orthogonal matrices prevent exploding or vanishing gradients, crucial in RNN training. This method facilitates stable training of RNNs by preserving gradient magnitude over time.

[Link to paper](#)

1.4 Milestones

To conclude the project proposal, several milestones will be stated for the group to complete before the determined dates.

12th April

- Introduction
- Find data and other resources to be used during our project.
- Initialization's impact analysis and begin LSTM initialization implementations

19th April

- Complete LSTM initializations' implementation and analysis
- Initial approach to RNN implementations